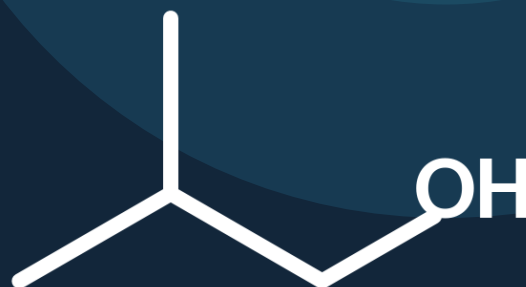


MARKET AND TECHNOLOGY ANALYSIS | SEPTEMBER 2026

Global Isobutanol Production

Chemistry, supply, demand, prices and policy:
a two-pass analysis of the world market



~0.8 Mt

global demand per year
(2023 estimate)

\$0.68-1.52B

published 2025 market-size
range

\$0.96-1.55/kg

Q1 2026 price, China to
India

~3 kt/yr

bio-isobutanol capacity
Gevo retains

01 | ABOUT THIS REPORT

Scope, method and how to read the numbers

This report examines global production of **isobutanol** (2-methyl-1-propanol, CAS 78-83-1): how it is made, who makes it, what it is used for, what it costs, and where policy and technology could change the picture. It reflects information available to **21 September 2026**, drawn from company filings and press releases, standards bodies, trade and specialist press, and syndicated market-research summaries.

The two-pass analysis

Pass 1: broad sweep	Pass 2: verification and stress test
About a dozen searches across market size, chemistry, producers, prices, policy and recent news. Goal: build a complete first picture and list every claim that needed testing.	A second round of targeted searches plus re-computation. Headline numbers were cross-checked against a second source or re-derived where possible. Where sources conflicted, primary sources (filings, standards bodies, government statements) won.

What the second pass changed

- **Bio-isobutanol scale.** Pass 1 syndicated reports implied sizeable bio-based production. Company filings show Gevo sold its Luverne ethanol plant (Nov 2025), kept about 1 million gallons a year of isobutanol capacity, and stopped advancing certain isobutanol projects (Aug 2026). The report now treats bio supply as sub-commercial.
- **Market size.** Seven publishers give 2025 values from \$0.68B to \$1.52B. Rather than pick one, pass 2 added a volume-times-price cross-check (page 8).
- **Latest developments.** Added India's June 2026 blend trials and the 16 September 2026 Praj-Gevo agreement, which post-date most syndicated reports.
- **Macro shock.** Added the 2026 Gulf conflict and Strait of Hormuz disruption after cross-reading C&EN and Chemistry World coverage.
- **Property data.** Flash point taken as 28 °C (ICSC-based) rather than a 37 °C supplier listing.

Reading guide

Source tiers. A: primary (company filings, standards bodies, government statements). B: trade and specialist press. C: syndicated market-research summaries, used for ranges only and never as sole support for a headline claim. "**Est.**" marks the author's own arithmetic, with assumptions stated. **Limits.** Paywalled reports were not accessed, isobutanol-specific capacity data are scarce, and several figures rest on a single source. This is analysis, not investment advice.

Contents

Page	Section	Page	Section
3	Executive summary	10	Prices and trade
4	The molecule: properties	11	Bio-isobutanol: promise and reality
5	Chemical route: the oxo process	12	Fuel and jet-fuel pathways
6	Bio-based and alternative routes	13	India spotlight
7	Demand and applications	14	Risks and scenarios to 2030
8	Market size: reconciling the numbers	15	Conclusions
9	Supply landscape and producers		

02 | EXECUTIVE SUMMARY

A small, oxo-linked chemical at a fuel-policy crossroads

Isobutanol is a mid-sized solvent and intermediate chemical whose supply is tied to propylene oxo plants. Its growth story now depends on fuels, chiefly India's diesel policy, but the bio-production capacity to serve that story does not yet exist at commercial scale.

- 1. A by-product-linked commodity.** Demand is about 0.8 Mt/yr and the chemical is made almost exclusively from propylene oxo chemistry. Supply follows the n:iso ratio operators choose in butyraldehyde plants.
- 2. Concentrated, integrated supply.** BASF, Eastman, Dow, OQ Chemicals, Mitsubishi Chemical, LG Chem and large Chinese complexes dominate. China is the biggest production base and the cheapest market.
- 3. Headline market sizes are unreliable.** 2025 estimates range from \$0.68B to \$1.52B. A volume-times-price check points to roughly \$0.8-1.0B for merchant chemical supply (Est.).
- 4. Prices are soft and regional.** The global average fell about 12.5% during 2025. Q1 2026 averages ran from \$0.96/kg in China to \$1.55/kg in India.
- 5. Bio-isobutanol remains marginal.** Gevo retains about 1M gal/yr, roughly 3 kt or 0.4% of demand (Est.), and dropped several isobutanol and jet-fuel projects in August 2026.
- 6. India could rewrite the demand curve.** Validation trials of a 2% isobutanol-diesel blend are under way and up to 15% is under discussion. Praj and Gevo signed an India commercialisation deal on 16 September 2026. A 2% blend needs about 1.8 Mt/yr (Est.), 2.3 times today's global demand.
- 7. Jet fuel favours ethanol.** ASTM allows isobutanol- and ethanol-derived alcohol-to-jet up to 50%, but Gevo's ATJ-30 project uses captive ethanol.
- 8. Shocks cut both ways.** The Gulf conflict, European oxo rationalisation and a global petrochemical glut pull supply and prices in different directions.

Key numbers at a glance

Metric	Value	Basis
Global isobutanol demand	~0.8 Mt/yr (2023)	Single consultancy estimate
Published market size, 2025	\$0.68B-\$1.52B	Seven publishers; median \$1.43B
Merchant price, Q1 2026	\$0.96-1.55/kg	China lowest, India highest
Route share	Almost all from propylene oxo	Hydrogenation of isobutyraldehyde
Retained bio-isobutanol capacity	~1M gal/yr (~3 kt)	Gevo, Luverne
India diesel demand, FY2025-26	94.7 Mt (29.7bn US gal)	2% blend needs ~1.8 Mt/yr (Est.)
Jet-fuel blend limit (ATJ-SPK)	Up to 50%	ASTM D7566 Annex A5

Bottom line

Treat isobutanol as an oxo by-product market until fuel demand is proven. The single largest swing factor is whether India moves from trials to a mandate, and whether anyone can build low-cost bio-isobutanol capacity fast enough to supply it.

03 | THE MOLECULE

Isobutanol: identity, properties and place among the butanols

Isobutanol (2-methylpropan-1-ol, $C_4H_{10}O$) is a colourless, flammable four-carbon alcohol. It is one of four butanol isomers (n-butanol, sec-butanol, tert-butanol and isobutanol), all industrially important. Its branched chain gives a lower boiling point than n-butanol (117.7 °C), and in most uses the two behave as close substitutes: Ullmann's notes they are often used interchangeably.



Figure 1. Skeletal structures of isobutanol and n-butanol.

Key physical and safety data

Property	Value	Property	Value
Molar mass	74.12 g/mol	Flash point	28 °C
Boiling point	107.9 °C	Autoignition temp.	415 °C
Melting point	-108 °C	Explosive limits	1.7-10.9 vol% in air
Density (liquid)	0.802 g/cm ³	Acute toxicity, LD50	2,460 mg/kg (rat, oral)
Vapour pressure	9 mmHg at 20 °C	NFPA 704	Health 1, Fire 3, Reactivity 0
Viscosity	3.95 cP at 20 °C	UN number	1212
Water solubility	8.7 mL per 100 mL	CAS / EC number	78-83-1 / 201-148-0

One supplier listing gives a flash point of 37 °C and a lower explosive limit of 2.4%; this report uses the ICSC-based values above as the more conservative safety figures.

Why the properties matter

- **As a solvent:** moderate polarity, a boiling point near 108 °C and slow evaporation suit it to lacquers, varnishes and as a starting point for esters such as isobutyl acetate.
- **As a fuel component:** compared with ethanol it has higher energy density, lower volatility and takes up far less water from the air, which is why Indian policymakers see it as blendable into diesel where ethanol separates.
- **Handling:** a flammable liquid (UN 1212) that can cause serious eye damage and skin and respiratory irritation, but among the least toxic butanols. Canada banned its use in cosmetics in 2009.

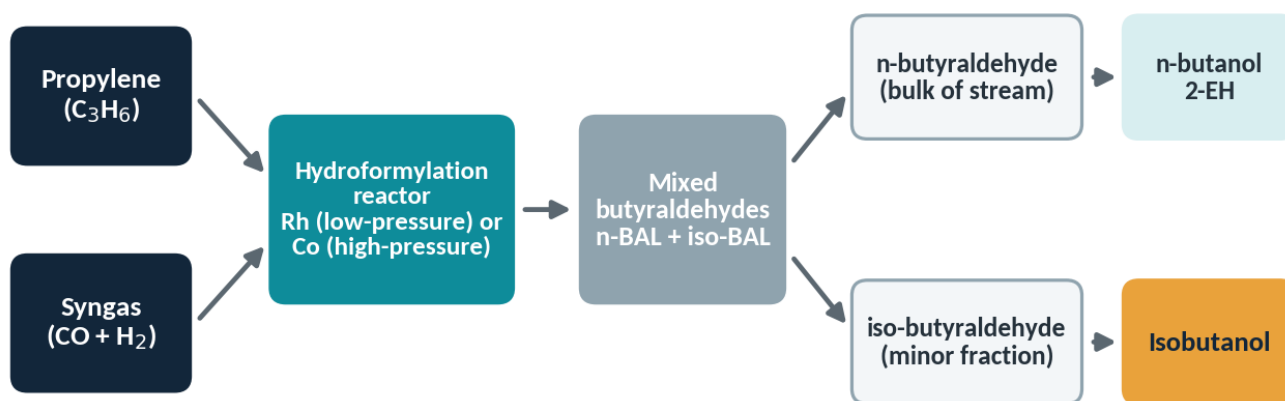
Why branching matters commercially

The iso-structure is what links isobutanol to the oxo process. Propylene adds CO at either end of the double bond: the terminal product becomes n-butyraldehyde and the branched product iso-butyraldehyde. Everything about isobutanol supply follows from that split (next page).

04 | CHEMICAL ROUTE

The oxo process: where almost all isobutanol comes from

Commercial isobutanol is made almost exclusively by hydrogenating isobutyraldehyde obtained from propylene via the oxo process. Propylene reacts with synthesis gas (CO and H₂) over a metal catalyst, giving a mixture of n- and iso-butyraldehyde. The aldehydes are separated and hydrogenated: the n-fraction becomes n-butanol (or 2-ethylhexanol), the iso-fraction becomes isobutanol.



hydrogenation then distillation; iso-butyraldehyde also feeds NPG and isobutyric acid

Figure 2. Simplified oxo route to butanols. Isobutanol is the minor branch of the butyraldehyde split.

The catalyst decides how much isobutanol exists

	Cobalt (high-pressure)	Rhodium/phosphine (low-pressure)
Catalyst	HCo(CO) ₄	HRh(CO)(PPh ₃) ₃
Conditions	110-170 °C, 1,500-4,000 psig	About 110 °C, 100-300 psig
n:iso ratio	2:1 to 4:1	8:1 to 12:1 (modern LP Oxo units 15:1 or higher)
Effect on isobutanol	More iso-aldehyde per tonne of propylene, but energy-intensive	Less iso-aldehyde; the shift to rhodium reduced isobutyraldehyde output
Operator control	Fixed by catalyst	Tunable: licensors such as Johnson Matthey offer ratio-tailoring technology

Scale cross-check (Est.)

World oxo-chemical consumption exceeds 10 Mt/yr, of which about 77% is propylene-derived n- and iso-butyraldehyde; butyraldehyde output is roughly 9-10 Mt/yr on an aldehyde basis. At n:iso ratios of 8:1 to 15:1 the iso fraction is about 6-11%, or 0.6-1.1 Mt/yr of isobutyraldehyde. That brackets the ~0.8 Mt/yr isobutanol demand estimate, once other iso-aldehyde outlets (neopentyl glycol, isobutyric acid) are allowed for.

Other chemical routes and cost drivers

- **Reppé carbonylation** gives the same products but uses the water-gas shift reaction to supply the hydrogen.
- **Guerbet condensation** of methanol with n-propanol can make isobutanol in principle; no commercial plant was identified in the sources reviewed.
- **Cost drivers:** propylene price and availability, syngas and energy costs, and the value of the larger n-butanol and 2-EH streams that share the plant. European operators also face high energy and carbon costs.

05 | ALTERNATIVE ROUTES

Bio-based and catalytic routes to isobutanol

Fermentation. Engineered microbes divert the valine biosynthesis pathway: α -ketoisovalerate is decarboxylated to isobutyraldehyde and reduced to isobutanol. UCLA work led by James Liao underpinned Gevo's 2008 exclusive licence, and Butamax began retrofitting Highwater Ethanol's plant in 2013. Because fermentation reuses ethanol-plant equipment, retrofits are cheaper than greenfield builds; one consultancy estimates about \$17M for 68 million litres of capacity.

Catalytic upgrading. Ethanol and methanol can be coupled over basic (hydrotalcite-type) catalysts at 150-500 °C. Ruthenium-diphosphine complexes have reached over 99% selectivity at over 75% conversion in the laboratory. Direct synthesis from syngas has given 11-14 wt% isobutanol in mixed products at 420 °C and 325 bar.



Figure 3. Simplified fermentation route to isobutanol.

Route comparison

Route	Feedstock	Maturity	Where it stands in 2026
Oxo, cobalt	Propylene, syngas	Mature	Legacy plants; higher iso yield but energy-intensive
Oxo, rhodium (LP Oxo)	Propylene, syngas	Mature, dominant in new plants	Lower iso fraction; ratio tunable
Reppe carbonylation	Propylene, CO, water	Mature, niche	Same products via water-gas shift
Fermentation (Bio-IBA)	Sugar, starch, molasses	Demonstration to first commercial	Gevo ~1M gal/yr retained; Praj demo plant for an Indian oil company
Guerbet / catalytic	Ethanol + methanol, or n-propanol	Laboratory and patents	High lab selectivity; no commercial plant identified
Syngas higher-alcohol	Syngas	Historic, laboratory	Isobutanol is a minor fraction of mixed alcohols

Feedstock options

- **Starch:** Gevo's Luverne plant used corn. **Cellulosic:** in 2016 Gevo made isobutanol from wood-waste sugars supplied by the Northwest Advanced Renewables Alliance.
- **India:** Praj and Gevo have spent over a decade adapting the technology to Indian sugar- and starch-based feedstocks, and industry experts propose sugarcane-molasses pilots.

The economic gap (Est.)

Merchant isobutanol at the Q1 2026 US average of \$1.06/kg equals about **\$3.2 per gallon** (1 gallon is about 3.04 kg). In 2016 Gevo targeted a variable production cost of \$3.00-3.50 per gallon at Luverne. That target is a decade old and excludes capital and overheads, but it shows the core problem: bio-isobutanol competes with a by-product chemical that already sells near its cost. Fuel-credit value, low-carbon premiums and mandates have to close the gap.

Retrofit logic

Fermentation appeals wherever surplus ethanol capacity exists. India reports ethanol distillery capacity of roughly 20 billion litres against a 10-11 billion litre blending need, which is why retrofits feature so strongly in Indian plans.

06 | DEMAND

What isobutanol is used for

Isobutanol demand is dominated by solvents and solvent-related derivatives, above all in paints and coatings. Ullmann's lists varnishes and ester precursors, such as isobutyl acetate, as the main applications, with isobutyl phthalate and adipate esters used as plasticisers.

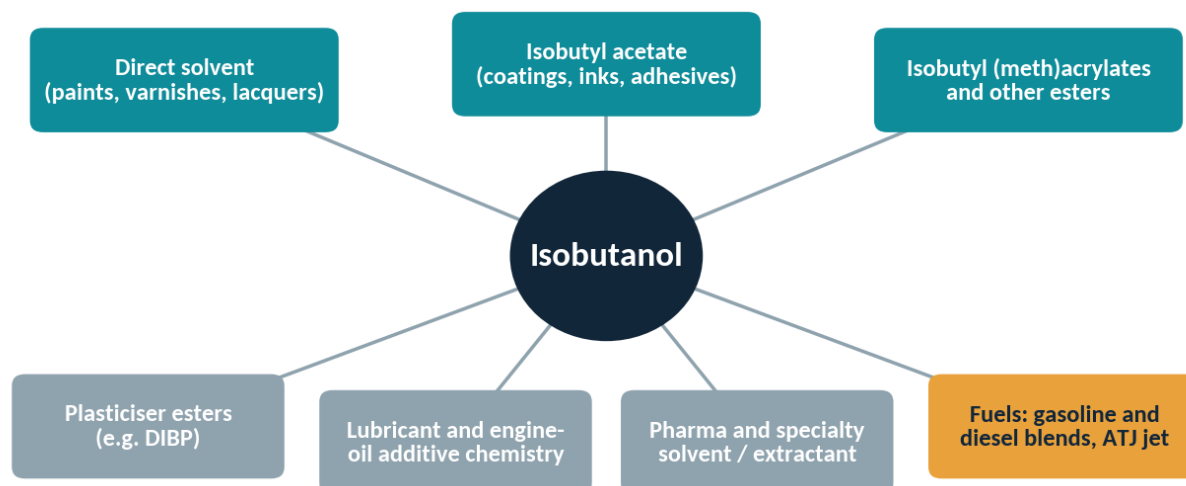


Figure 4. Main outlets for isobutanol. Fuel uses (amber) are policy-driven and still small.

Use	How isobutanol is used	End markets	Notes
Solvent and coalescent	Direct solvent in lacquers and varnishes; film-forming aid in waterborne paint	Architectural and industrial coatings	One syndicated source cites 2-4 wt% in waterborne latex paint
Isobutyl acetate	Esterification with acetic acid	Coatings, adhesives, inks	Growth linked to low-VOC reformulation
Plasticiser esters	Isobutyl esters of phthalic and adipic acids (e.g. DIBP)	Plastics, rubber, dispersions	About 68% of India's demand in FY2022; phthalate phase-down in Europe
Acrylates and methacrylates	Isobutyl acrylate and methacrylate monomers	Resins, coatings, adhesives	-
Lubricant and cleaning chemistry	Anti-wear and anti-corrosion engine-oil components; polishing and cleaning	Automotive	-
Pharma and specialty	Solvent, extractant and intermediate	Pharma, fine chemicals	-
Fuels	Gasoline blendstock, diesel blend, jet-fuel feedstock	Transport	See pages 12-13

Regional pattern

Asia-Pacific has historically been the largest region, with as much as 42% of demand in 2019; China, India, Japan and South Korea lead consumption. Mature North American and European demand is projected to grow at roughly 1.5-3.5% a year through 2030, against 3-5% for Asia-Pacific.

Demand drivers

Coatings and construction recovery; low-VOC reformulation; fuel-substitution policy, especially in India.

Headwinds

Weak construction and coatings demand in 2025; phthalate phase-down in Europe; easy substitution with n-butanol.

07 | MARKET SIZE

Reconciling the numbers

Published market sizes for 2025 range more than two-fold. Volume times price is a better anchor than any single headline.

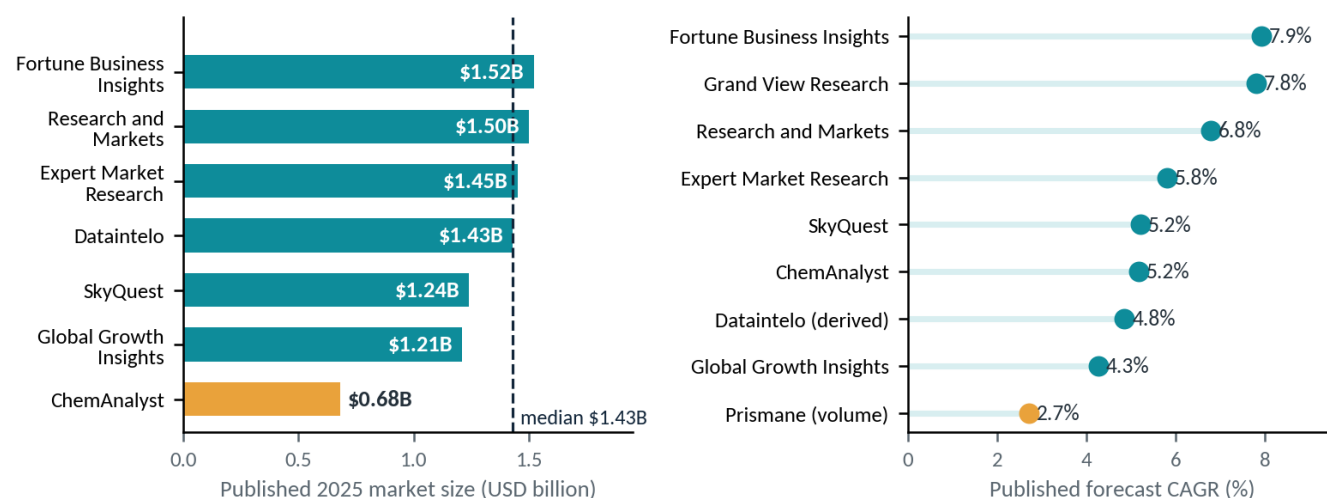


Figure 5. Left: published 2025 market size by publisher. Right: published forecast growth rates (the amber point is a volume rate, and Dataintelo's rate is derived from its 2025 and 2034 values).

Estimate register

Publisher	2025 value	Forecast	CAGR
Fortune Business Insights	\$1.52B	\$3.01B by 2034	7.91%
Research and Markets	\$1.50B	\$2.08B by 2030	6.78%
Expert Market Research	\$1.45B	\$2.55B by 2035	5.80%
Dataintelo	\$1.43B	\$2.19B by 2034	4.85% (derived)
SkyQuest	\$1.24B	\$1.86B by 2033	5.2%
Global Growth Insights	\$1.21B	\$1.84B by 2035	4.26%
ChemAnalyst	\$0.68B	\$1.19B by 2036	5.16%
Grand View Research / Prisma	\$1.28B (2023) / 800 kt (2023)	n/a	7.8% (2024-30) / 2.7% volume (2024-36)

Triangulation (Est.)

The median published 2025 value is \$1.43B and the mean \$1.29B. But demand of about 0.8 Mt/yr at \$1.0-1.2/kg implies only **\$0.8-1.0B**. The higher estimates probably use a broader scope, such as fuel-grade or bio-based volumes, or different price bases. One publisher assigns 40.6% of 2025 value to bio-based isobutanol and another more than 63% in 2020. That is hard to square with Gevo's roughly 1M gal/yr of retained capacity, so such shares should be treated as red flags.

Growth: what the spread implies

Value CAGRs run from 4.3% to 7.9%, while the one volume forecast is 2.7%. Value can outgrow volume only through higher prices or a richer mix, and 2025 prices fell. For planning, a volume-led base case of about 2.5-4% a year is more defensible (author's judgement) unless fuel demand materialises.

How to use these figures

Quote a range, name the scope, and prefer volume and price. Where a report attributes large shares to bio-based supply, ask which plants produce it.

08 | SUPPLY

Who makes isobutanol, and where

Isobutanol supply sits inside integrated oxo complexes. Producers are rarely isobutanol specialists: the chemical is a co-product of n-butanol and 2-ethylhexanol chains, often consumed captively in derivatives such as isobutyl acrylate.

Region	Notable producers	Notes
Europe	BASF (Ludwigshafen), OQ Chemicals (Oberhausen), Grupa Azoty, Perstorp	Rationalisation: Dow closed its Terneuzen cracker in phases, BASF cut Verbund rates, LyondellBasell ran Wesseling at reduced rates. OQ raised iso- and n-butanol prices by \$110/t in July 2024.
North America	Eastman (Longview, Texas), Dow (US Gulf Coast), OQ Chemicals Corp. (Bay City, Texas)	Lower feedstock costs than Asia or Europe. Gevo adds ~1M gal/yr of bio capacity.
China	Luxi Chemical (~40 kt/yr), Chengzhi (~50 kt/yr), Guangxi Huayi (~27 kt/yr); Wanhua and Sinopec in butyraldehyde	Largest producer; ample capacity keeps China the cheapest market. Capacities are approximate.
Other Asia	Mitsubishi Chemical, KH Neochem (Japan); LG Chem, Hanwha (Korea); Dairen (~20 kt/yr), Nan Ya, Chang Chun (Taiwan); BASF PETRONAS (Malaysia)	Japanese suppliers JNC and KH Neochem raised prices in April 2025 on tight supply and feedstock costs.
India	Andhra Petrochemicals (~3 kt/yr); BPCL and Indian Oil n-butanol units yield isobutanol as co-product	Import-dependent, hence the highest prices. Merchant demand about 67 kt in 2024.
Middle East and Africa	SABIC, Tasnee; Sadara, Equate; Sasol (South Africa)	Gulf exports disrupted since the Hormuz closure; large inflows from the Middle East and US into Asia in Q3 2025.

Capacity figures for Chinese and Taiwanese plants come from a syndicated summary and are approximate. No independent, complete isobutanol capacity database was accessible.

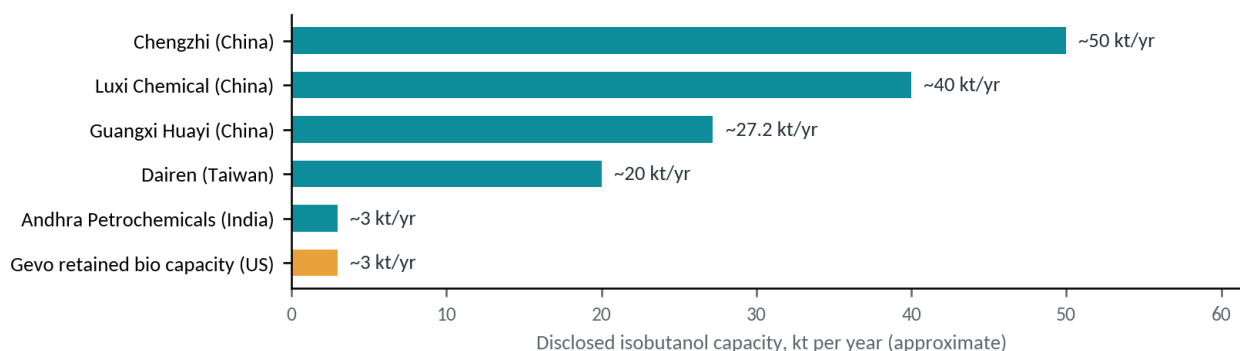


Figure 6. Disclosed single-site isobutanol capacities. Gevo's figure is 1M gal/yr converted at 3.04 kg/gal (Est.).

Structural features of supply

- **Co-product economics.** Volumes follow n-butanol and 2-EH demand and the n:iso ratio chosen at each plant, so isobutanol capacity does not respond directly to isobutanol prices.
- **Small unit scales.** Disclosed single-site isobutanol capacities range from about 3 kt to about 50 kt a year, tiny beside the ~9-10 Mt/yr butyraldehyde system.
- **Regional imbalance.** Asia is long and exports; Europe has been shrinking; India imports. Trade flows follow those gaps.
- **Industry backdrop.** Chemical makers faced a global capacity glut and the deepest downturn in a generation even before the 2026 conflict.

Implication

Pricing power sits with a handful of oxo majors, while Chinese scale sets the global price floor. New isobutanol supply will arrive mainly as a by-product of butanol expansions, not as dedicated isobutanol plants, unless a fuel mandate changes the economics.

09 | PRICES AND TRADE

Soft prices, wide regional gaps

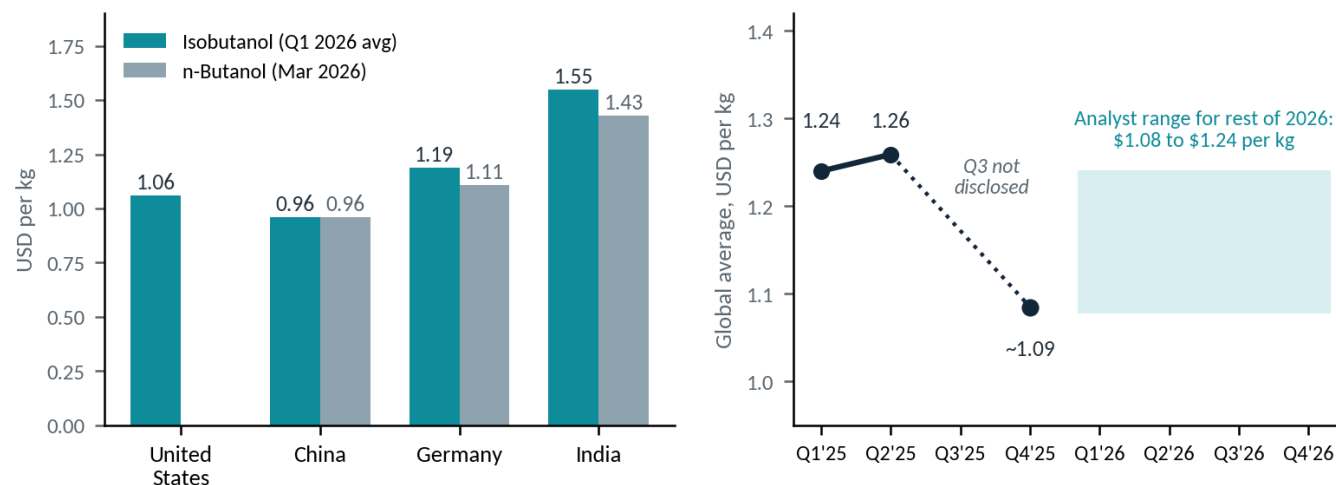


Figure 7. Left: Q1 2026 isobutanol averages against March 2026 n-butanol (different providers and periods, so indicative). Right: global average isobutanol price.

Region	Q1 2026 average	Q2 2026 range	Comment
United States	\$1.06/kg	\$1.03-1.09	Up 12.8% on Q4 2025 as restocking met higher propylene-linked costs
China	\$0.96/kg	\$0.93-0.99	Cheapest region; ample domestic capacity
Germany	\$1.19/kg	\$1.16-1.22	High energy and compliance costs keep prices firm
India	\$1.55/kg	not disclosed	Highest price; import dependence

A price of \$1.06/kg equals about \$1,060/t or \$3.2/gal (Est.).

What moved prices

- **2025 weakness.** The global average fell from \$1.24/kg in Q1 to about \$1.09 in Q4, roughly 12.5%, on weak construction and coatings demand, ample supply and soft propylene. In Q3 Asia saw port congestion and large inflows from the Middle East and US.
- **Isobutanol against n-butanol.** Isobutanol sat at parity in China and about 7% above n-butanol in Germany and about 8% above in India (Est.).
- **Upstream tightness, little visible pass-through.** European butyraldehyde rose 52% year on year in Q1 2026 on cracker closures and carbon costs, but the isobutanol series reviewed shows no comparable jump (Germany \$1.19/kg in Q1 2026). Both come from one publisher, so treat the contrast with caution.
- **Outlook.** Analysts expected a \$1.08-1.24/kg global range for the rest of 2026, assuming a gradual coatings and construction recovery and ample Asian supply. Q3 2026 prints were not available.

Trade signals

India extended anti-dumping duties on n-butanol imports for five years in April 2026, after finding that exporters from Malaysia, South Africa and the US were still dumping and that Chinese duties on similar goods could divert supply towards India. Because isobutanol is a butanol-plant co-product, such rulings plausibly shape the economics of Indian butanol units and their isobutanol output (author's inference).

Watch

India's premium (isobutanol \$1.55/kg against \$0.96/kg in China) is an import-substitution signal and a plausible motive for interest in domestic supply (author's inference).

10 | BIO-ISOBUTANOL

Promise and reality

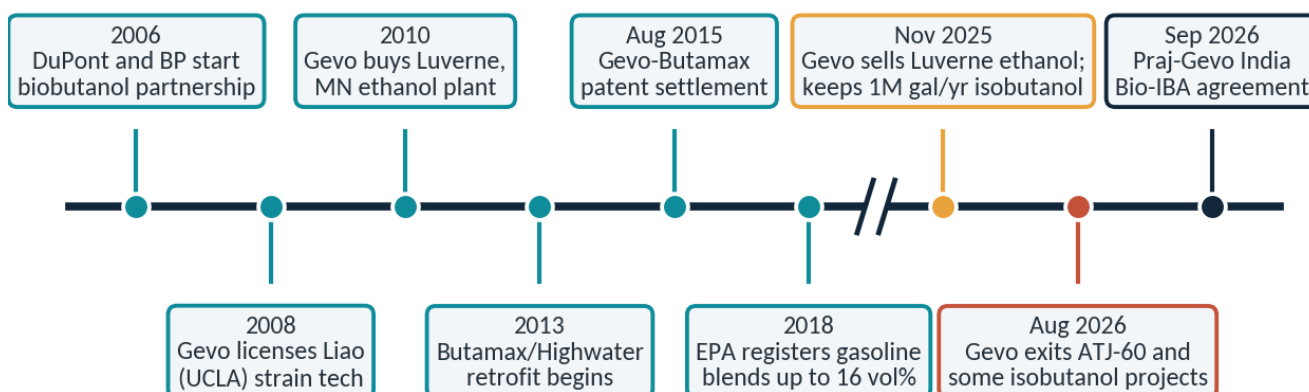


Figure 8. Selected milestones in bio-isobutanol.

How it unfolded

DuPont and BP formed a biobutanol partnership in 2006, and Gevo licensed UCLA's strain technology in 2008 and bought its Luverne, Minnesota ethanol plant in 2010. A four-year patent fight between Gevo and Butamax ended in August 2015 with a cross-licence in which both could sell up to 30 million gallons a year royalty-free. In 2018 the US EPA registered Butamax isobutanol blends of up to 16 vol% in gasoline.

Expectation against outcome

In 2015 Gevo had expected to be producing 500 million gallons of isobutanol a year at multiple sites by then, and had accumulated a \$325M deficit. In November 2025 it sold the Luverne ethanol business and kept isobutanol assets for about 1 million gallons a year. In August 2026, after a strategic review, it decided to focus on its North Dakota ethanol platform and a 30M gal/yr ATJ project, stopped advancing ATJ-60, other ATJ projects and certain isobutanol projects, and booked a \$176M non-cash impairment. It had earlier withdrawn from the DOE loan-guarantee process for ATJ-30.

Where bio-isobutanol is still alive

- **India.** Praj holds exclusive rights to deploy Gevo's Bio-IBA technology in India, initially for diesel blending, and is building India's first commercial demonstration plant for an oil marketing company.
- **Niche uses.** Gevo's retained Luverne assets target chemicals, racing-fuel feedstock, gasoline and jet fuel, and Praj has demonstrated integrated jet-fuel production from both isobutanol and ethanol.

Why it stalled (author's assessment)

- **Price ceiling.** Oxo isobutanol sells near \$3.2/gal (Est.), close to the cost targets bio producers quoted a decade ago.
- **Policy dependence.** The bio case rests on fuel credits and mandates rather than solvent demand, and no isobutanol-specific mandate appeared in the sources reviewed.
- **Capital allocation.** Gevo's returns now come from low-carbon ethanol, carbon capture and tax-credit monetisation, not isobutanol.

Scale check (Est.)

Gevo's retained capacity of 1M gal/yr is about 3,000 t (1M gal x 3.785 L x 0.802 kg/L), or roughly 0.4% of the ~0.8 Mt/yr global market. Bio-isobutanol is a technology option today, not a supply source.

11 | FUELS AND JET FUEL

Three fuel outlets, three different stages of readiness

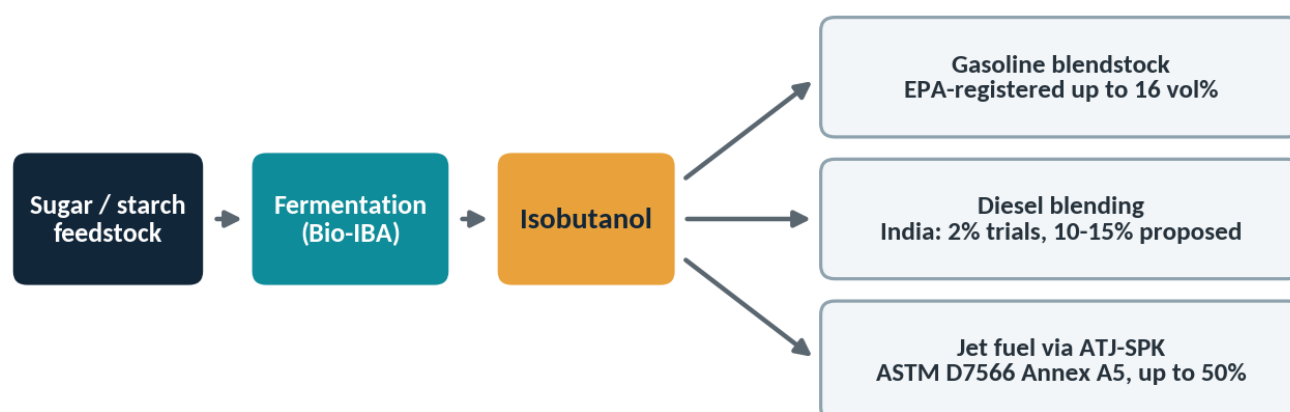


Figure 9. Fuel outlets for isobutanol.

Outlet	Status, September 2026	Volume signal	Main blockers
Gasoline blendstock	EPA registered blends up to 16 vol% in 2018	No volumes found in the sources reviewed	Cost against ethanol and gasoline; limited supply
Diesel blending (India)	2% validation trials since June 2026; 10-15% under consideration	1.8 Mt/yr (2%) to 13.5 Mt/yr (15%) (Est.)	No formal notification; vehicle compatibility and warranty; cost; supply
Jet fuel (ATJ-SPK)	ASTM D7566 Annex A5: isobutanol first, ethanol added in 2018; up to 50% blend	ATJ-30 (30M gal/yr) in development at Gevo North Dakota	Project financing; Gevo left the DOE loan-guarantee process
Bio-based chemicals	Isobutanol gives a route to bio-based p-xylene and terephthalic acid	No commercial plant identified	Cost against fossil aromatics

Why diesel, and why isobutanol

Ethanol cannot be blended into diesel because of poor solubility and phase separation, and Indian trials of ethanol-diesel blends failed. Isobutanol mixes far better with diesel, is less corrosive and takes up less water. Diesel is also the larger prize: it is roughly 40% of India's petroleum-product use and nearly double petrol consumption.

Jet fuel: optionality rather than base case

The alcohol-to-jet standard originally admitted only isobutanol, and ethanol was added in 2018; ICAO lists ethanol, isobutanol and isobutene as feedstocks. Since then developers have gravitated to ethanol because it is abundant: Gevo's ATJ-30 relies on captive ethanol from its North Dakota plant, and Praj validated ethanol-to-jet with Axens technology in January 2026 after first proving isobutanol. Isobutanol's jet-fuel role therefore looks like a technology option, not a demand pillar.

Reading across

On the numbers above, only the Indian diesel proposal is large enough to move global isobutanol volumes. Gasoline registration and jet-fuel approval are enabling steps that have not, so far, produced offtake.

12 | INDIA SPOTLIGHT

From blend trials to a possible mandate

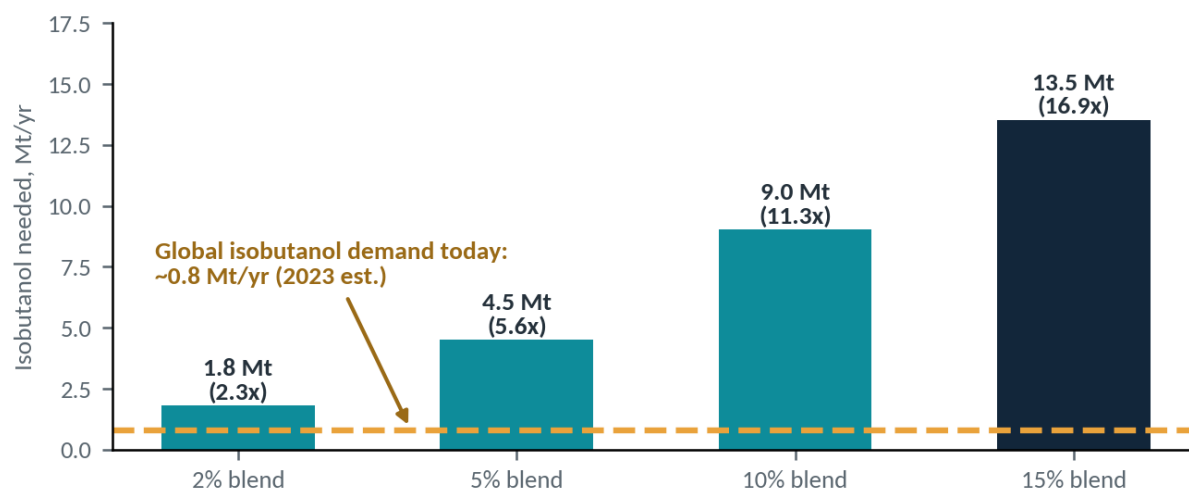


Figure 10. Isobutanol needed for volumetric blends in Indian diesel (Est.): 29.7 billion US gallons (94.7 Mt) of high-speed diesel in FY2025-26, converted at 3.785 L/gal and 0.802 kg/L, against ~0.8 Mt/yr global demand.

Date (2026)	Event
17 Feb	Praj plans a semi-commercial demonstration unit within 12-18 months; ARAI trialling a 10% blend
5 Mar	Oil marketing companies cannot blend beyond limits without a mandate; industry seeks a policy signal and possible subsidy
31 Mar	Praj's indigenous technology tested by BPCL, ARAI and engine makers; ~20 bn litres of ethanol capacity against 10-11 bn litres of blending need
Late May	Minister Gadkari says a 15% blend is being evaluated; on 29 May the MoRTH secretary says a mandate is likely before end-2026 after encouraging BPCL results
26 Jun	Validation trials of a 2% blend begin with MoRTH, HPCL, ARAI, Praj and Tata Motors
Early Jul	Gadkari repeats that India is working towards allowing up to 15% isobutanol in diesel by 2027
16 Sep	Praj and Gevo sign an India development and commercialisation agreement; Praj is building a commercial demonstration plant for an oil marketing company

The scale problem (Est.)

India's merchant isobutanol market was about 67 kt in 2024, forecast to reach about 89 kt, with roughly 68% going to plasticisers. A 2% diesel blend would need about 1.8 Mt/yr, some 27 times the entire Indian merchant market and 2.3 times the world's. Linear scaling of the quoted retrofit cost (about \$17M per 68 million litres) to the 2.25 billion litres a 2% blend needs implies roughly **\$0.5-0.6B** of capex, before feedstock, logistics and any new build. This is an order-of-magnitude figure, not a project estimate.

Open questions

- Will a formal notification be issued, and with what blend specification? None had been issued in the coverage reviewed (latest July 2026).
- Will vehicle makers give warranty guidance? Early trials show no significant wear at 10%, but no maker has issued a compatibility statement.
- Which feedstock: grain, broken rice or sugar-industry molasses, and who invests without a viability-linked subsidy?

13 | RISKS AND SCENARIOS

What could move isobutanol to 2030

Macro backdrop: the 2026 Gulf conflict

The Strait of Hormuz has been effectively closed to non-military shipping since 4 March 2026, stranding oil, gas and chemical exports from Iran, Iraq, Kuwait, Bahrain, Saudi Arabia, the UAE and Qatar. Brent crude surged above \$120 a barrel at one point, and C&EN judged it had upended petrochemical expectations for 2026. Asian chemical makers now face constrained feedstock and rerouted supply, while low-cost North American producers may gain. For isobutanol, published price series through Q2 2026 moved only modestly, but Q3 data were not available. Higher crude also strengthens India's case for fuel substitution.

Risk and driver register

Factor	Direction	Horizon	Why
Propylene cost and availability	Two-way	Near term	European cuts tighten supply; Asian supply ample
Chinese overcapacity and exports	Bearish	Ongoing	Cheapest region; duties elsewhere can divert supply
European rationalisation	Bullish (regional)	Near term	Butyraldehyde +52% year on year in Q1 2026
Gulf disruption	Two-way	Near term	Stranded exports and feedstock stress against a higher oil price
Coatings and construction demand	Two-way	Cyclical	Weak in 2025
India diesel mandate	Strongly bullish if enacted	2026-2030	1.8-13.5 Mt/yr (Est.); needs new capacity
Bio-isobutanol economics	Gate on the mandate	2027 onward	Gevo retreat; Praj demo plant
Jet-fuel policy	Mild	2027 onward	Ethanol pathway favoured
VOC and phthalate rules	Mixed	Ongoing	Helps low-VOC solvents; hurts DIBP

Scenarios to 2030 (author's judgement, not forecasts)

Scenario	Assumptions	Global demand, 2030	Price signal
Base	Organic growth of 2.7-3.5% a year from 0.8 Mt in 2023; no diesel mandate at scale	About 0.96-1.02 Mt	\$1.0-1.2/kg; regional gaps persist
India upside	Mandate enacted; part of the diesel pool reaches a 2% blend; retrofit-led Bio-IBA capacity built	About 1.0 Mt organic plus 0.5-1.8 Mt of fuel demand	Fuel-linked premiums; bio route viable with support
Downside	Mandate delayed or diluted; weak construction; Chinese export push	About 0.85-0.9 Mt	Below \$1.0/kg; margins squeezed

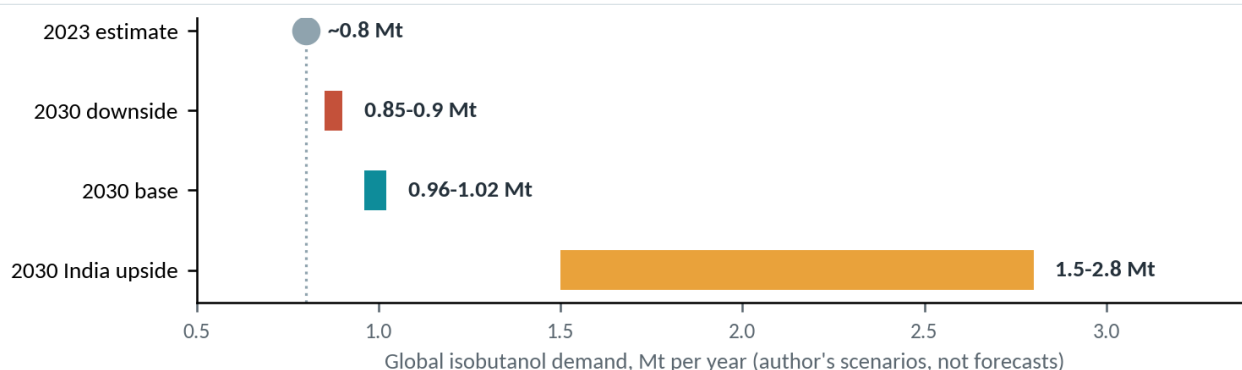


Figure 11. Scenario ranges for 2030 global isobutanol demand (author's judgement).

Base-case arithmetic: $0.8 \text{ Mt} \times 1.027^7 = 0.96 \text{ Mt}$ and $0.8 \text{ Mt} \times 1.035^7 = 1.02 \text{ Mt}$. Upside fuel demand is 25-100% of the 2% blend requirement of 1.8 Mt. Scenarios illustrate sensitivities and carry no probabilities.

14 | CONCLUSIONS

What to take away, and what to watch

Conclusions

- **Model it as an oxo by-product.** Supply follows butyraldehyde plants; prices follow propylene, Chinese scale and coatings demand.
- **Anchor on volume times price.** Market-size headlines differ by more than 2x; ~0.8 Mt at \$1.0-1.2/kg gives \$0.8-1.0B (Est.).
- **Bio-isobutanol is an option, not a supply.** Gevo retains ~1M gal/yr and has retreated from several projects.
- **India is the swing factor.** A 2% diesel blend would exceed today's global demand; delivery depends on policy and unbuilt capacity.

What to watch in the next 12 months

- A formal Indian notification and blend specification.
- Commissioning of Praj's oil-company demonstration plant.
- Q3 and Q4 2026 price prints after the Gulf disruption.
- European oxo capacity decisions.
- Gevo's ATJ-30 financing decision.

Data gaps

No independent capacity database was accessible; the 0.8 Mt volume rests on a single estimate; Q3 2026 prices and Indian Q2 prices were unavailable.

How this report was put together

The analysis was built in two passes. The first was a broad sweep across chemistry, producers, prices, policy and recent news to build a complete first picture. The second was a targeted round of verification: every headline number was cross-checked against an independent source or re-derived from first principles, and where evidence conflicted, primary material such as company filings, standards bodies and government statements took precedence over trade press and syndicated market-research summaries. Figures marked "Est." throughout are the author's own arithmetic, with the underlying assumption stated alongside each one.

Abbreviations used in this report

Term	Meaning	Term	Meaning
ATJ	Alcohol-to-jet (fuel pathway)	MoRTH	India's Ministry of Road Transport and Highways
ATJ-SPK	ATJ synthetic paraffinic kerosene	ARAI	Automotive Research Association of India
Bio-IBA	Bio-based isobutanol	BPCL / HPCL	Bharat Petroleum / Hindustan Petroleum
LP Oxo	Low-pressure (rhodium-catalysed) oxo process	EPA	US Environmental Protection Agency
DIBP	Diisobutyl phthalate	ASTM	ASTM International, a standards body
NPG	Neopentyl glycol	ICAO	International Civil Aviation Organization
CAGR	Compound annual growth rate	NFPA 704	US fire/health/reactivity hazard rating
kt / Mt	Kilotonnes / megatonnes (1 Mt = 1,000 kt)	ICSC	International Chemical Safety Card

Final word

Isobutanol is a small, by-product chemical today, priced and supplied on the coattails of propylene oxo plants. Whether it becomes a materially larger market by 2030 turns on one thing more than any other: whether India converts its diesel-blend trials into an enforceable mandate, and whether bio-isobutanol capacity can be built fast enough, and cheaply enough, to supply it.